



NYC Traffic Collision Frequency & Time-Based Patterns

Analyze collision patterns across months, weekdays and hours to identify peak accident period

Time Analysis

Location & Causes

Severity Analysis

Total Accidents



238K

Peak Month



Mar

Peak Day



Friday

Peak Hour



16

Filter here →

Year



All



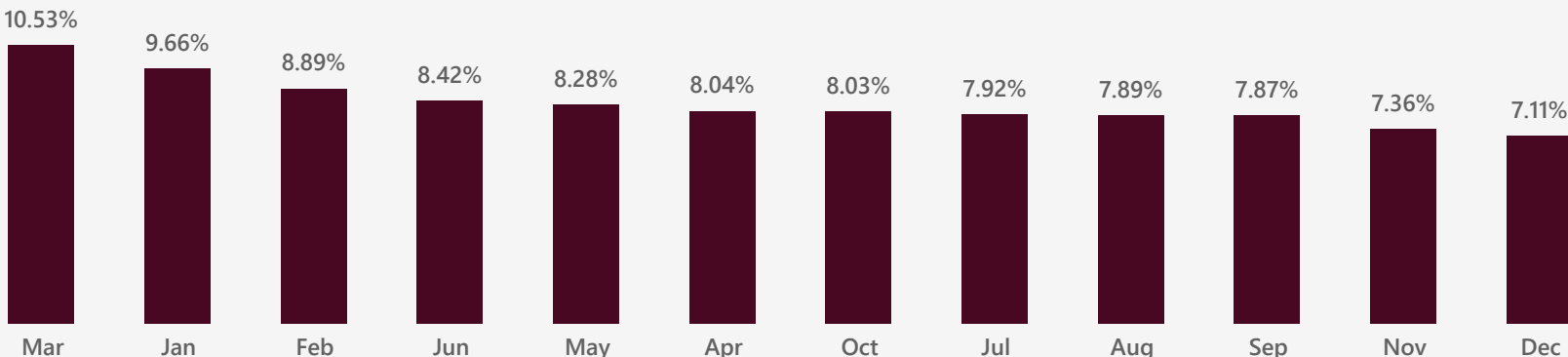
Month



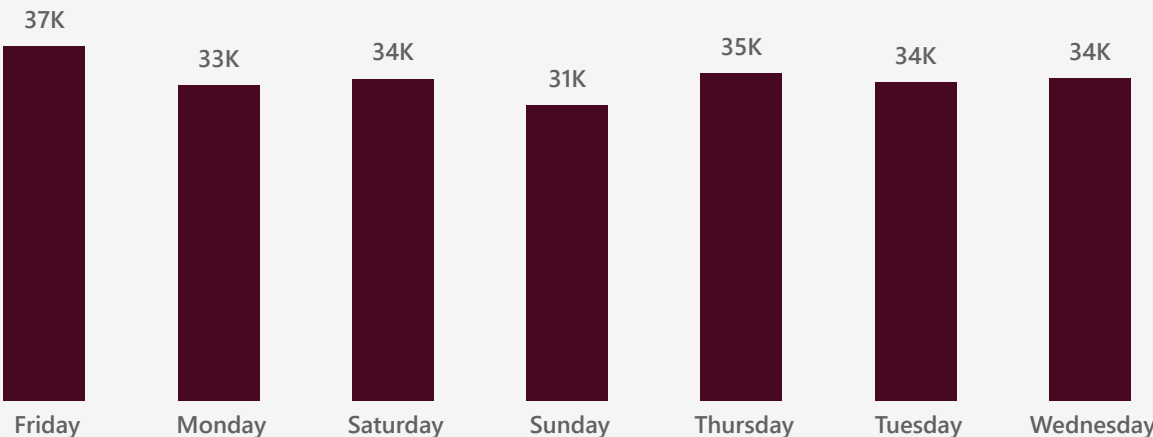
All



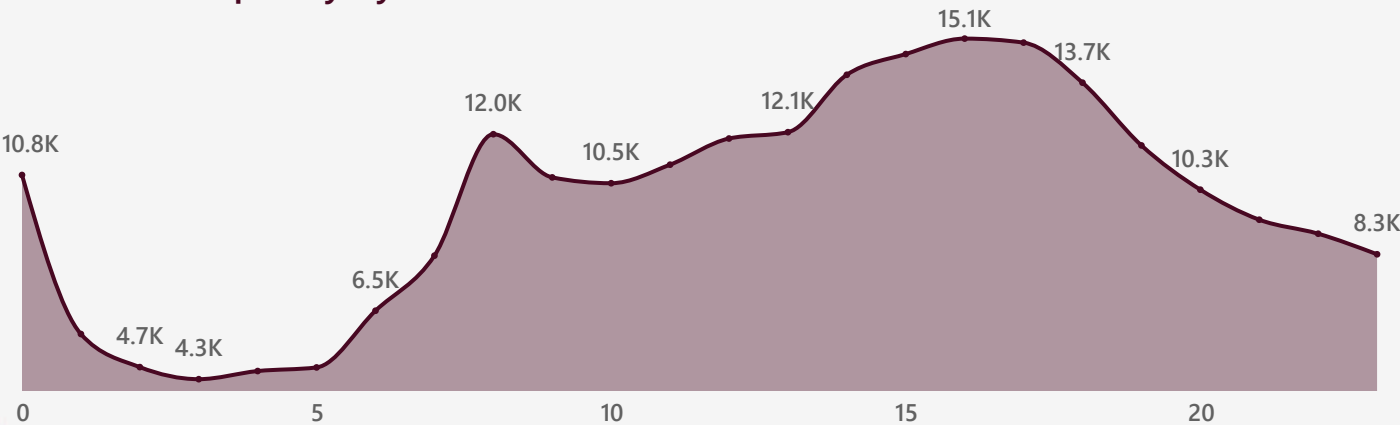
Monthly Percentage of Total Collisions



Collision Frequency by Day of Week



Collision Frequency by Hour



Insights

March recorded the highest percentage of collisions (10.53% of all accidents). After March, the number of collisions gradually reduced towards the end of the year, with November (7.36%) and December (7.11%) recording the lowest percentages. Collisions happened most often on Fridays (37K cases) and peaked around 4:00 PM, showing that Friday evening rush hour is the busiest collision period.

Recommendations

Increase traffic monitoring and enforcement on Fridays, especially between 3:00 PM and 6:00 PM when collisions are highest. Road safety campaigns should also be introduced before March to help reduce the increase in collisions during that period.

High-Risk Collision Locations & Contributing Factors

Identify collision hotspots and the primary contributing factors to support targeted traffic safety interventions.

Time Analysis

Location & Causes

Severity Analysis



Top Street



Belt Parkway

Top Street %

%

1.56%

Contributing Factor



Driver Inattention/Distracted

Fatal Collisions



613

Filter here

Year

Borough

All

All

Top 5 Streets by Collision Count

Belt Parkway

3.7K

Broadway

2.8K

Atlantic Avenue

2.2K

Long Island Expressway

2.2K

Brooklyn Queens Expressway

2.2K

Leading Factors in Fatal Collisions

Unspecified

175

Unsafe Speed

130

Driver Inattention/Distracted

74

Failure to Yield Right-of-Way

46

Traffic Control Disregarded

33

Leading Contributing Factors

Driver Inattention/Distracted

58K

Unspecified

58K

Failure to Yield Right-of-Way

17K

Following Too Closely

16K

Passing or Lane Usage Improper

11K

Insights

Belt Parkway recorded the highest collision count, accounting for 1.56% of all 238K reported collisions. Driver Inattention/Distracted was the leading cause overall (58K cases), while Unspecified causes (175) and Unsafe Speed (130) were the leading causes of fatal collisions. This shows that the biggest cause of collisions is different from the biggest cause of fatal collisions.

Recommendations

Belt Parkway should be given priority for road safety reviews, including traffic management, road signs and enforcement. Since speeding is more common in fatal collisions than distraction, safety campaigns should focus on both speeding and distracted driving instead of treating them as the same problem.

Collision Severity and Human Impact Assessment

Evaluate injuries, fatalities, and high-risk locations to understand the overall impact of traffic collisions across New York City.

Time Analysis

Location & Causes

Severity Analysis

Filter here →

Year

All

Vehicle Type

All

Persons Injured

116K

Persons killed

635

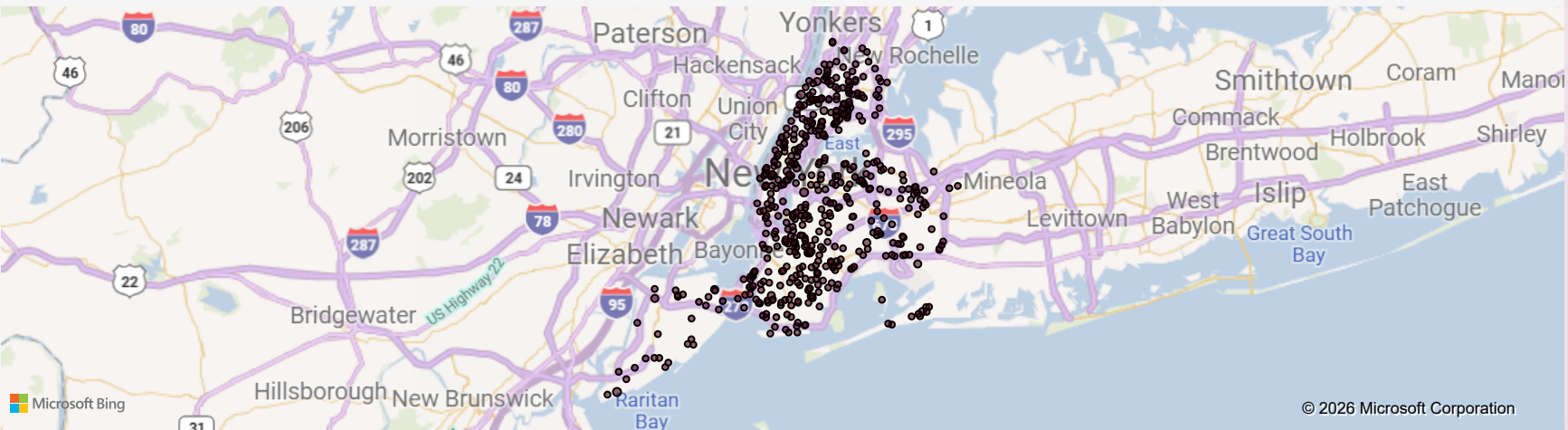
Injury Rate

48.7%

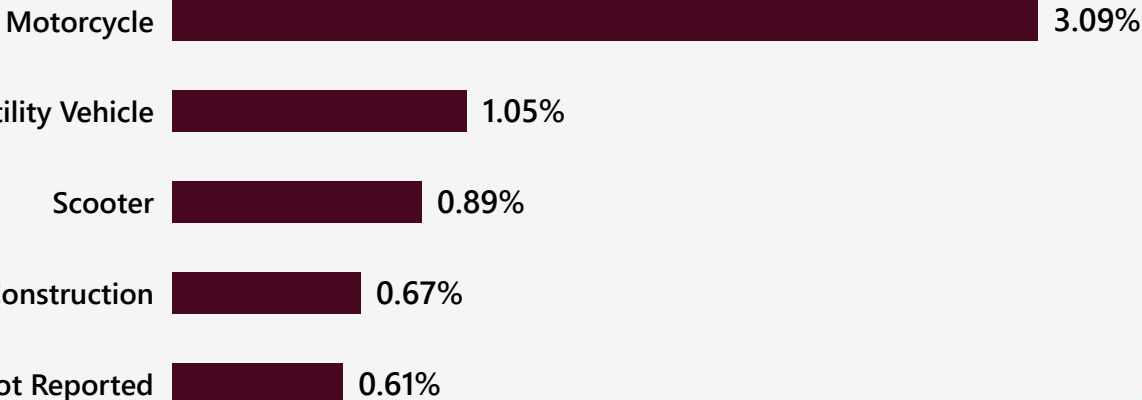
Fatality Rate

0.27%

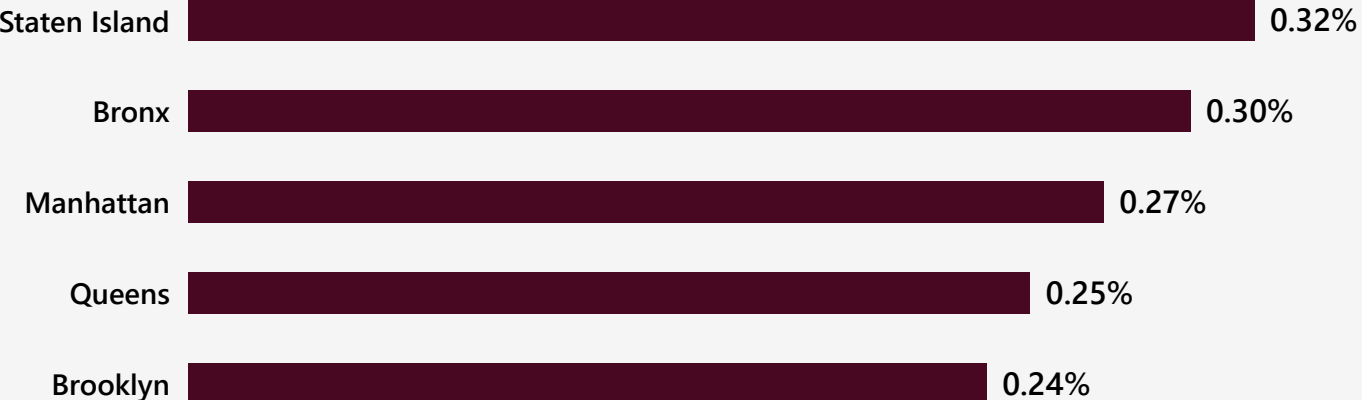
Collision Severity Hotspot



Fatality Rates Across Vehicle Types



Borough-Level Fatality Rates



Insights

Out of 238K reported collisions, 116K (48.7%) resulted in injuries, but only 635 were fatal (0.27%). Despite the low citywide fatality rate, motorcycles recorded the highest fatality rate (3.09%), almost three times that of Utility Vehicles (1.05%) and over eleven times the city average. Staten Island also had the highest borough fatality rate (0.32%).

Recommendations

Motorcycle safety should be treated as a separate road safety priority because of its much higher fatality risk. The high fatality rate in Staten Island should also be looked into further so that the factors behind these severe collisions can be identified and addressed.